DESIGN BASIS FOR DATA CENTER FACILITIES

Baseline Design Basis Methodology for The Hyperscale Facility Class

Electric Sector Facility Design Basis Series

First Facility-Class Application of the Electric Sector Design Basis Methodology

Version 1.1

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For review by engineers, asset owners, regulators, and insurers.

Parent series: Electric Sector Design Basis Framework (DBA-ES) — emerging

Revision History

Version	Date	Author	Description
0.1	April 2026	T. Roxey	Initial working draft. Released for technical review. Established document architecture, all nine sections, five postulated event categories, three-tier consequence taxonomy, acceptance criteria framework, gap analysis framework, and military action extension bridge.
0.2	April 2026	T. Roxey	Incorporated Uptime Institute Tier Standard reference. Added compliance floor language in Section 6.1, establishing Tier IV certification as the minimum assumed baseline for hyperscale facilities. Added Section 7.4 positioning Tier Certification as the starting point for gap analysis—no structural changes to Sections 1 through 5 or Sections 8 and 9.
1.0	April 2026	T. Roxey	First complete release version. Added Section 2.4 formally establishing the four distinguishing characteristics of the hyperscale facility class as the governing analytical pillars of the DC series. Strengthened acceptance criteria in Section 6 with bounded thresholds where engineering judgment supports specification. Converted Section 9 open items to Deferred Items with formal resolution pathways. Built as a proper series instrument, replacing the v0.2 working text. Released for series integration.
1.1	May 2026	T. Roxey	Section 10 added: Facility Class Instruments, comprising Section 10.1 (Inheritance Model) and Section 10.2 (Facility Class Register). Register uses ONG naming convention: FC1 (Hyperscale, this document, Active), FC2 (Colocation), FC3 (Enterprise), FC4 (Edge) — all Not Yet Addressed. Full DC series rename to ONG convention deferred to dedicated revision pass.

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Series: DBA-DC Data Center and AI Infrastructure Series

Position in hierarchy: L1 Baseline Design Basis — foundational document; all other series documents inherit from this instrument.

Section 1 — Purpose and Scope

1.1 Purpose

This document establishes the Design Basis for the data center facility class. It is the first application of a facility-class design-basis methodology to a major electric-sector facility outside the nuclear sector. No equivalent document currently exists. The data center sector — like large hydroelectric facilities, high-voltage direct current interconnects, and major natural gas compressor stations — operates without a formal design basis. That absence is the condition this document begins to correct.

A design basis, in the tradition established by nuclear safety engineering over six decades, is the set of postulated conditions that a facility must be demonstrated to withstand. It defines the envelope of credible events, the required system responses, and the acceptance criteria against which actual facility capability is measured. Everything that follows from a design basis — engineering specifications, safety system requirements, capital investment decisions, regulatory compliance obligations — flows from the rigor and scope of that foundational definition.

For nuclear facilities, the design basis and the regulatory framework built around it have driven decades of engineering investment and safety improvement. The electric sector has relied instead on operational standards and procedural frameworks — instruments that tell facility operators what processes to follow, but not what physical conditions their facilities must be capable of withstanding. This document introduces the missing layer.

1.2 Scope

This document addresses hyperscale data center facilities as the initial facility class within the electric sector design basis series. Hyperscale facilities — defined here as data center campuses exceeding 100 megawatts of information technology load capacity — are the highest-consequence tier in the data center sector. They present the greatest concentration of digital infrastructure, the deepest dependency footprint on other critical infrastructure sectors, and the most significant grid-interaction characteristics. The design basis for lower-consequence facility tiers — colocation, enterprise, and edge — is intended as a subsequent extension of this framework.

The geographic scope is national, with principal reference to the United States domestic hyperscale clusters. The Northern Virginia cluster — the largest concentration of data center capacity in the world — is the primary reference case throughout this document. The methodology is transferable to allied-nation hyperscale clusters without modification; jurisdictional annexes addressing non-domestic regulatory environments are anticipated but not yet drafted.

1.3 What This Document Is Not

This document is not a threat assessment. It does not characterize specific adversaries, current intelligence regarding threat actors, or the probability of any particular attack scenario. Those analyses are performed by the DBA-DC military action extension series and the associated use case documents, which derive their analytical authority from this design basis.

This document is not a policy argument. The regulatory gap in data center physical security — the complete absence of mandatory physical security standards for commercial data center facilities — is documented in the DBA-DC Policy Companion (DBA-DC-POL-001). This document assumes that gap as context but does not argue for its closure. Its purpose is to supply the analytical instrument that makes a mandatory standard meaningful: without a design basis, a mandatory standard has no floor against which to measure compliance.

This document is not a military action analysis. The extension of this design basis to military action initiating conditions is the subject of the DBA-DC military action extension series, which applies the four DBA-MA differentiators — intentionality, persistence, precision, and denial — to the postulated events defined here. Section 8 of this document establishes the bridge between the baseline design basis and that extension.

1.4 Relationship to the Electric Sector Design Basis Series

This document is the first facility-class application of a design basis methodology that does not yet have a formal parent document at the sector level. The electric sector Design Basis Series parent — to be designated DBA-ES — will establish the baseline methodology at sector scale: what a design basis means for electric sector facilities, how the consequence taxonomy is structured, how gap analysis is conducted, and what the acceptance criteria framework looks like across facility classes.

That parent document does not exist yet. It will emerge inductively from this document and from the facility-class design bases that follow it — for large hydroelectric facilities, high-voltage direct current interconnects, and major natural gas compressor stations. This is the same developmental path by which the nuclear sector's design-basis methodology evolved: facility-specific analyses accumulated until the pattern was sufficiently established to be codified at the sector level. Section 3 of this document is written with that future codification in mind. The methodology layer in Section 3 is expressed in transferable terms, with data-center-specific content clearly separated from the general framework, so that the parent document can be assembled by lifting Section 3 across facility classes without reconstruction.

Section 2 — Facility Class Definition

2.1 Defining the Hyperscale Facility Class

For purposes of this design basis, a hyperscale data center facility is defined by three characteristics, all of which must be present.

Scale. The facility must exceed one hundred megawatts of information technology load capacity. This threshold marks the boundary at which a data center campus becomes a significant load on the bulk electric system — comparable in grid impact to a large industrial customer or a mid-sized generating unit — and at which the campus architecture introduces simultaneity vulnerability: shared power delivery, cooling infrastructure, and network connectivity across multiple buildings means that disruption of common infrastructure elements produces cascading effects across the entire campus simultaneously.

Campus architecture. Hyperscale facilities characteristically operate as campuses: multiple large buildings on contiguous or proximate land parcels, sharing utility service entrance infrastructure, substation equipment, cooling water systems, and network interconnection points. This architecture is an efficiency characteristic in normal operations and a concentration vulnerability under stress conditions.

Multi-sector dependency. The facility's customer base must span multiple critical infrastructure sectors or include government and military cloud service customers. Facilities serving a single commercial sector may have significant consequences if disrupted. Still, the design-basis threshold is reached when a facility disruption propagates across the sectoral boundaries of the critical infrastructure framework. Facilities hosting federal government cloud environments, military logistics and decision-support systems, financial sector operational technology, or the control systems of other regulated critical infrastructure sectors satisfy this criterion.

2.2 Principal Domestic Hyperscale Clusters

Four domestic hyperscale clusters meet the facility-class definition. Its geographic footprint, grid interaction profile, and dependency consequences characterize each.

Northern Virginia. The Loudoun and Prince William County corridor is the world's largest concentration of data center capacity. Estimates of its share of total domestic data center capacity range from 25% to 35%. The cluster hosts primary Amazon Web Services, Microsoft Azure, and Google Cloud regions serving the eastern United States, the federal government, and the financial sector. Its grid interaction with the PJM Interconnection is material: Northern Virginia's data center load growth has driven substation investment and transmission planning in PJM's service territory. The cluster's concentration is its defining vulnerability characteristic: a coordinated event affecting shared infrastructure — power delivery, transmission corridors, cooling water supply — could affect the majority of the cluster simultaneously.

Central Ohio. The Columbus metropolitan area hosts major Amazon Web Services and Google Cloud campuses that serve the central United States and provide redundancy for East Coast operations. The Ohio cluster is the primary domestic geographic alternative to Northern Virginia for federal government cloud workloads.

Phoenix metropolitan area. The Phoenix cluster hosts major AWS, Microsoft, and Google campuses and serves as the primary West Coast-adjacent concentration. Its geographic separation from Northern Virginia makes it the primary domestic redundancy location for East Coast hyperscale infrastructure. Its grid interaction is with the Western Electricity Coordinating Council service territory.

Chicago. The Chicago cluster hosts significant colocation and hyperscale capacity and serves as a primary network interconnection hub for the central United States. Its grid interaction is with both the PJM and MISO service territories.

2.3 Grid Interaction Profile of the Hyperscale Facility Class

The hyperscale facility class exhibits a grid interaction profile that distinguishes it from every other large industrial load category, and this profile is central to the consequence taxonomy developed in Section 5.

Magnitude. Individual hyperscale campuses consume power at a scale that places them among the largest single electricity customers in their utility service territories. The Northern Virginia cluster as a whole represents a load concentration that materially affects PJM transmission planning. The design basis must account for this magnitude in both the facility-level and grid-level consequence categories.

Load behavior under stress. Data center load can drop rapidly and completely under stress conditions — protective shutdown, thermal management response, or deliberate workload migration — in ways that are invisible to grid operators because they occur behind the customer meter. The April 7, 2015, Washington DC area disturbance documented by NERC established this mechanism precisely: load disappeared from the grid because customer-owned behind-the-meter protective equipment operated in response to a voltage excursion, and the grid operator had no visibility into the initiating mechanism. A hyperscale campus undergoing a controlled shutdown presents the same observability gap to a greater extent.

Workload portability. The hyperscale facility class is unique among large electricity consumers in that its productive output — computing workload — is portable. A manufacturing facility that loses power loses production; it does not redirect it to another facility. A hyperscale data center that loses power or anticipates a threat condition can migrate its computing workloads to a geographically remote facility through software orchestration commands operating at the IP layer. That migration is a load dropout at the originating node and a load surge at the receiving node, occurring simultaneously, at a rate that grid planning systems are not designed to anticipate or manage.

This workload portability characteristic is the mechanism that creates the cross-sector cascading consequence addressed in Section 5.3. It is the first design basis in any critical infrastructure sector to incorporate this mechanism as a postulated event category formally.

2.4 Four Distinguishing Characteristics of the Hyperscale Facility Class

The hyperscale data center facility class exhibits four characteristics that, taken together, distinguish it from every other large electric-sector facility and govern the analytical approach for this design basis and all subordinate documents in the series. Every document in the DC series addresses the analytical implications of all four. No document in the series is complete if it does not account for all four characteristics as governing conditions.

Concentration. Hyperscale capacity is organized into geographic clusters — the Northern Virginia corridor, the Central Ohio corridor, the Phoenix metropolitan area — where multiple campus operators share utility service infrastructure, transmission corridors, and network interconnection points. This clustering is an economic efficiency characteristic under normal operating conditions and a simultaneity vulnerability under stress conditions. An event that affects shared infrastructure — a transmission corridor, a water supply, a power substation serving multiple campuses — affects the entire cluster simultaneously, not merely the individual facility that might otherwise bear the consequence. No prior facility class in the electric sector presents this cluster-scale simultaneity characteristic.

Foundational Dependency. The hyperscale facility class is not one critical infrastructure sector among many — it is the substrate on which all other critical infrastructure sectors operate. Financial sector clearing and settlement, grid control systems, pipeline SCADA, transportation logistics, military command and logistics support, and emergency communications all depend on hyperscale computing infrastructure as a prerequisite for function. This foundational dependency means that disruption of the hyperscale facility class propagates horizontally across the critical infrastructure framework in a manner that no other single-sector failure does. The three-tier consequence taxonomy in Section 5 — and particularly the Tier 3 cross-sector cascading category — is a direct expression of this characteristic.

Regulatory Vacuum. The hyperscale facility class operates without mandatory physical security standards. The electric sector's mandatory reliability framework — NERC CIP — stops at the customer meter. The data center sector's security posture is governed entirely by voluntary frameworks, operator judgment, and contractual obligations to individual customers. The Cybersecurity and Infrastructure Security Agency's CIPAC partnership structure was terminated in March 2025. No successor mechanism with enforcement authority exists. This regulatory vacuum means that the design basis in this document has no current mandatory compliance pathway — it establishes the analytical standard that a mandatory framework would require, not a compliance obligation that currently exists. The policy argument for closing that gap is developed in the DBA-DC Policy Companion.

Foreign Ownership. A material portion of commercial data center capacity serving United States government and military cloud customers is owned or operated by entities with ownership structures that include adversary-affiliated investment. This characteristic creates a pre-positioning threat vector that is analytically distinct from the kinetic and cyber threats addressed in the postulated event categories: foreign-owned or foreign-influenced infrastructure may have been engineered, operated, or configured in ways that serve adversary interests at the moment of activation. The foreign ownership characteristic is analyzed in depth in DBA-DC-L3-003; this design basis notes it as a distinguishing characteristic of the facility class because it affects the adversary model that governs the military action extension analysis in Section 8.

Section 3 — Design Basis Methodology

This section is the transferable layer of this document. It establishes what a design basis means for an electric sector facility, how postulated events are identified and bounded, what the consequence taxonomy structure looks like, and how gap analysis is conducted. Readers of subsequent facility-class design bases — for large hydroelectric facilities, HVDC interconnects, or natural gas compressor stations — will find this section reproduced in substance in those documents, with facility-specific content substituted in Sections 2, 4, 5, and 6.

3.1 The Design Basis Concept Applied to Electric Sector Facilities

In nuclear safety engineering, the design basis is the set of postulated accidents that a facility must be demonstrated to withstand. The design basis defines the envelope of credible events — the conditions the facility could plausibly encounter over its operational life — and establishes the safety system performance requirements and acceptance criteria that bound those events. The design basis does not assert that these events will occur; it asserts that if they occur, the facility must be capable of the specified response.

Transposed to electric sector facilities, the design basis concept requires three adaptations.

First, the consequence taxonomy must be extended beyond the facility boundary. Nuclear design basis accidents are analyzed primarily for their radiological consequences, which are contained by a hierarchy of physical barriers. Electric sector facilities lack equivalent physical containment. Their consequences propagate outward through the electric grid, dependent infrastructure systems, and the downstream services those systems support. The design basis for an electric sector facility must formally account for consequences that cross the facility fence line.

Second, the postulated event categories must reflect the specific physics and operational characteristics of the facility class. Nuclear design-basis accidents are organized around the reactor coolant pressure boundary, reactivity control, and decay heat removal — the three safety functions that determine the severity of nuclear consequences. For a hyperscale data center, the analogous organizing functions are power continuity, thermal management, and connectivity. Postulated events are those that challenge one or more of these functions in a manner that, if unmitigated, leads to a defined consequence category.

Third, the acceptance criteria must be expressed in terms of operational states, not just physical parameters. Nuclear acceptance criteria are expressed in physical terms: peak cladding temperature, maximum reactor coolant pressure, and containment integrity. Electric sector acceptance criteria must include operational state criteria: facility availability, workload continuity, grid contribution, and service restoration timeline. These criteria have engineering content, as well as operational and commercial dimensions that nuclear analysis does not address.

3.2 Postulated Event Identification

Postulated events are identified through a structured process that considers the facility's physical systems, its operational dependencies, and the external conditions it may encounter. The identification process is not bounded by historical frequency — a postulated event need not have occurred previously to be included in the design basis. It must be credible, in the sense that a reasonable analyst applying established engineering judgment would identify it as within the range of conditions the facility may encounter, and it must be consequential, in the sense that its occurrence challenges one or more of the facility's critical functions in a manner that could lead to a defined consequence.

For the hyperscale facility class, postulated events are organized into five categories, developed in Section 4: power supply events, thermal management events, connectivity events, physical integrity events, and workload migration events. The last category — workload migration — is novel to this design basis and has no analog in prior facility-class design bases in any sector. Its inclusion reflects the workload portability characteristic of the hyperscale facility class described in Section 2.3.

Code	Category	Event	Governing Condition
PE-1	Power Supply	Short-duration utility loss	Up to 15 minutes; bridging capacity required
PE-2	Power Supply	Extended-duration utility loss	15 minutes to 72 hours; sustained generation required
PE-3	Power Supply	Catastrophic-duration utility loss	Beyond 72 hours; design basis boundary condition
PE-4	Power Supply	Voltage excursion	Sustained excursion outside ANSI C84.1 Range A
TM-1	Thermal	Loss of mechanical cooling	Primary chiller or cooling tower failure
TM-2	Thermal	Loss of cooling water supply	Municipal or on-site storage disruption
TM-3	Thermal	Combined power and thermal stress	Simultaneous degradation of both systems
CE-1	Connectivity	Single-provider loss	One ISP or carrier path is lost
CE-2	Connectivity	Area-wide connectivity loss	All fiber ingress/egress paths severed
PI-1	Physical Integrity	Fire	In-facility fire activating suppression
PI-2	Physical Integrity	Physical damage to critical points	Cause-agnostic; MA extension is cause-specific

Code	Category	Event	Governing Condition
WM-1	Workload Migration	Deliberate migration under threat	Operator-commanded IP-layer migration
WM-2	Workload Migration	Automated failover	System-triggered migration without deliberate command
WM-3	Workload Migration	Cross-interconnection load transient	Cluster-level aggregate grid consequence

3.3 Consequence Taxonomy Structure

Consequences are organized in three tiers, reflecting the hierarchy from facility-level to cross-sector cascading effects. The three-tier structure is the transferable element of this taxonomy; the specific consequence categories within each tier are facility-class-specific and are defined in Section 5.

Tier	Designation	Consequence Scope
Tier 1	Facility-Level	Computing capacity loss, data integrity consequences, physical infrastructure damage, personnel safety consequences. Contained within the facility boundary.
Tier 2	Grid-Level	Load dropout effects, frequency perturbation, voltage excursion propagating from the facility to the bulk electric system.
Tier 3	Cross-Sector Cascading	Consequences propagating from grid-level effects into dependent critical infrastructure sectors and government, military, and commercial services. Formally defined innovation of this design basis.

3.4 Gap Analysis Framework

The gap analysis is the instrument that converts design basis findings into capital investment decisions. It is the mechanism by which the design basis produces actionable outcomes for asset owners, regulators, and insurers.

The gap analysis process compares a specific facility's demonstrated capabilities against the acceptance criteria established in Section 6 for each postulated event category. A gap exists where the facility cannot demonstrate that it meets an acceptance criterion. Gaps are characterized by category, severity, and remediation pathway.

Gap characterization by category identifies which postulated event type the gap affects and which tier of the consequence taxonomy is implicated. A power supply gap that affects only Tier 1 consequences represents a different risk profile and a different capital priority than a power supply gap whose consequences cascade to Tier 3.

Gap characterization by severity establishes the magnitude of the shortfall between demonstrated capability and acceptance criterion. Severity characterization is the basis for prioritization: not all gaps can be closed simultaneously, and the design basis must support a prioritization framework that asset owners and regulators can apply consistently.

Remediation pathway characterization identifies the engineering measures available to close the gap, their cost, their timeline, and their interdependencies with other facility systems. This characterization is the direct input to capital project planning.

Section 4 — Postulated Events

4.1 Category 1 — Power Supply Events

Power supply events are the most consequential single category for the hyperscale facility class because power disruption initiates a time-bounded cascade through backup systems that, if uninterrupted, terminates in facility-wide shutdown. The design basis must address the full range of power supply disruption scenarios, from momentary voltage excursions through extended utility outage.

PE-1: Loss of Utility Power, Short Duration (less than fifteen minutes)

Initiating condition: loss of utility power supply to the facility service entrance due to grid disturbance, equipment failure, or physical damage to transmission or distribution infrastructure. Duration: up to fifteen minutes. Expected facility response: automatic transfer to uninterruptible power supply bridging capacity and backup generation. This scenario is within the design envelope of well-maintained facilities and establishes the baseline against which backup system adequacy is measured.

PE-2: Loss of Utility Power, Extended Duration (fifteen minutes to seventy-two hours)

Initiating condition: same as PE-1, but the utility power supply is not restored within the bridging capacity of installed backup systems. Duration: fifteen minutes to seventy-two hours. Expected facility response: sustained operation on backup generation, with fuel consumption at planned load. The design basis challenge in this scenario is fuel supply: a hyperscale campus operating at full load on backup generation can consume diesel fuel at a rate that exhausts on-site storage within hours. Fuel resupply under conditions of area-wide disruption — road damage, competing demand, supply chain stress — is the postulated stress condition.

PE-3: Loss of Utility Power, Catastrophic Duration (beyond seventy-two hours)

Initiating condition: physical destruction or sustained denial of access to utility transmission or distribution infrastructure serving the facility, such that restoration within seventy-two hours is not credible. This scenario represents the design basis boundary for the power supply category. It is included not because it is a normal reliability event but because it is the condition that the military action extension is specifically designed to analyze. The design basis must establish what the facility can and cannot sustain under this condition, so that the military action extension has a defined baseline from which to assess military action initiating conditions.

PE-4: Voltage Excursion Without Loss of Power

Initiating condition: sustained voltage deviation on the utility supply, short of complete power loss, sufficient to cause behind-the-meter protective equipment to operate. ANSI C84.1 establishes Range A service voltage as plus or minus five percent of nominal; excursions beyond this range, sustained for intervals sufficient to activate protective relay settings, constitute this postulated event. The April 7, 2015 Washington DC area disturbance is the reference case: a sustained fifty-eight-second fault caused customer-owned protective equipment to operate, dropping approximately 532 megawatts from the PJM system. A hyperscale campus experiencing the same initiating condition would shed load through behind-the-meter protection in a manner invisible to the grid operator.

4.2 Category 2 — Thermal Management Events

Thermal management events challenge the facility's ability to maintain computing equipment within operating temperature ranges. The time constant of thermal management events is shorter than power supply events: loss of cooling under high-load conditions can cause protective temperature shutdowns within minutes, requiring faster automated response than any power supply scenario.

TM-1: Loss of Mechanical Cooling

Initiating condition: failure or loss of chiller plant, cooling tower, or primary mechanical cooling system. Duration to first protective equipment shutdown: dependent on facility thermal mass, ambient conditions, and operating load. This is the most time-critical postulated event in the thermal management category. The design basis thermal dwell time — the interval between cooling loss and the first server protective shutdown — establishes the minimum response window for cooling system restoration or controlled load reduction. This interval is the governing design requirement for cooling system redundancy and transition time.

TM-2: Loss of Cooling Water Supply

Initiating condition: interruption of the municipal water supply or on-site water storage system that sustains evaporative cooling. Many hyperscale facilities are significant water consumers; the Northern Virginia cluster has materially affected local water planning. Disruption of water supply — through infrastructure damage, area-wide emergency diversion, or deliberate denial — affects cooling system capacity independently of electrical power availability. A facility may retain full electrical power and backup generation capability while being unable to sustain thermal management under full computing load due to water supply loss.

TM-3: Combined Power and Thermal Stress

Initiating condition: simultaneous degradation of both power supply and cooling capacity. This scenario tests the facility's ability to manage load reduction in a coordinated manner that prevents both electrical overload of backup generation and thermal overload of reduced cooling capacity. It is more demanding than either single-system scenario because the mitigation actions for power stress and thermal stress may conflict: reducing computing load reduces heat generation but also reduces the urgency of fuel conservation; maintaining thermal management under backup power consumes generator capacity that could extend operational duration. The governing design challenge is a coordinated load reduction protocol that optimizes both thermal tolerance and backup power duration simultaneously.

4.3 Category 3 — Connectivity Events

Connectivity events challenge the facility's ability to send and receive data from its customers and from the broader internet. A facility that retains power and thermal management but loses external connectivity has lost its productive function, even if its physical infrastructure is intact. Connectivity events may be independent of power supply events or may accompany them in coordinated attack scenarios.

CE-1: Loss of Network Connectivity, Single Provider

Initiating condition: disruption of a single internet service provider's or network carrier's connectivity to the facility. This is the baseline connectivity scenario. A facility with adequate provider diversity can sustain productive function during CE-1 by routing traffic through alternative providers. CE-1 establishes the design basis requirement for multi-provider diversity and the minimum acceptable failover time.

CE-2: Loss of Network Connectivity, Area-Wide

Initiating condition: disruption of all network connectivity to the facility through damage to the physical fiber and cable infrastructure serving the facility campus. This scenario is credible for any facility served by a limited number of physical ingress and egress pathways for fiber infrastructure, which is the case for several Northern Virginia campus locations where fiber routes share physical corridors. CE-2 is the connectivity equivalent of PE-3: a condition that exceeds the redundancy provided by multiple providers and that no current commercial connectivity architecture can sustain indefinitely without physical restoration.

4.4 Category 4 — Physical Integrity Events

Physical integrity events challenge the structural and mechanical integrity of the facility itself, independent of the electrical and connectivity systems it depends on.

PI-1: Fire

Initiating condition: fire originating within the facility — in electrical equipment, cooling systems, or the building structure — that activates suppression systems and may require evacuation and temporary shutdown of affected areas. Fire suppression activity, whether water-based, chemical, or gaseous agent, can itself cause equipment damage and extended outage beyond that caused by the fire itself. PI-1 is the facility-class analog of the nuclear sector's loss-of-coolant accident in one respect: the mitigation action and the initiating event both contribute to the consequence.

PI-2: Physical Damage to Critical Infrastructure Points

Initiating condition: physical damage to facility-level critical infrastructure — utility service entrance, substation equipment, backup generator plant, fuel storage, or primary cooling infrastructure — from any cause. This category is facility-class agnostic with respect to cause; the design basis establishes the consequence regardless of whether the cause is equipment failure, severe weather, or deliberate attack. The military action extension addresses cause-specific analysis for deliberate targeting. PI-2 establishes the requirement for critical infrastructure point characterization, which is the direct input to that targeting analysis.

4.5 Category 5 — Workload Migration Events

Workload migration events are unique to the hyperscale facility class and have no analog in prior design bases for any critical infrastructure facility. They arise from the workload portability characteristic described in Section 2.3: the facility's productive output can be redirected to geographically remote facilities through software orchestration commands operating at IP layer.

WM-1: Deliberate Workload Migration Under Threat Condition

Initiating condition: the facility operator, in response to an actual or anticipated threat to the facility, executes a software-commanded migration of computing workloads from the facility to one or more geographically remote receiving facilities. The migration command operates at IP layer and does not require physical switching actions, coordination with grid operators, or advance notification to any regulatory body. The migration takes effect within minutes.

Facility-level consequence: rapid and near-complete dropout of the facility's information technology load from the local grid. At Northern Virginia scale, this represents a potential load dropout measurable in hundreds of megawatts to multiple gigawatts, occurring in minutes.

Receiving facility consequence: simultaneous load surge at the receiving facility's grid interconnection point, in the same magnitude as the originating facility's dropout.

Grid consequence: the dropout and the surge occur at nodes in different grid interconnections — PJM at the originating facility and, depending on the receiving location, ERCOT or WECC at the receiving facility. The two grid events are causally linked through the IP orchestration layer but are not observable as linked by grid operators in either interconnection. Each grid operator sees an unexplained anomaly in their own system without visibility into the cross-interconnection mechanism.

WM-2: Automated Failover Under Disruption Condition

Initiating condition: automated workload migration triggered by loss of connectivity, power disruption, or other facility-level event that activates the facility operator's failover logic without deliberate human command. This scenario is analytically distinct from WM-1 because it may occur with less operator awareness and with less advance staging of receiving facility capacity. The grid consequences are the same in character but potentially less orderly in execution, since the deliberate preparation that characterizes WM-1 — pre-staging receiving capacity, sequencing migration order — may not be present.

WM-3: Cross-Interconnection Load Transient

This is the aggregated grid consequence scenario. It does not describe a single facility event but the combined grid effect of WM-1 or WM-2 at the cluster level. If a threat condition or disruption event affects the Northern Virginia cluster broadly — through area-wide power disruption, area-wide physical damage, or area-wide threat perception that triggers multiple operators' migration logic simultaneously — the aggregate load dropout in the PJM Eastern Interconnection could be of sufficient magnitude to cause a measurable frequency excursion.

The mechanics of this consequence are established in the NERC disturbance record. The September 18, 2007 MRO disturbance demonstrated that a sudden separation event producing overgeneration relative to load caused an overfrequency excursion to 60.87 Hz, with cascading relay operations at remote nodes set to factory overfrequency trip thresholds. The cross-interconnection load transient scenario represents the deliberate or forced analog of that event, initiated from the data center sector rather than from transmission system dynamics.

The receiving interconnection — most consequentially ERCOT given its electrical isolation from the Eastern Interconnection — faces a mirror-image challenge. A multi-gigawatt load surge arriving from compute migration, at a rate that outpaces ERCOT's automatic generation control response, presents a frequency stability challenge that ERCOT's planning framework has not evaluated for this initiating condition. The acceptance criterion for this scenario and the resolution pathway for the engagement it requires are addressed in Section 6.4 and Deferred Item DI-1 in Section 9.

Section 5 — Consequence Taxonomy

5.1 Tier 1 — Facility-Level Consequences

Tier 1 consequences occur within the facility boundary. They are organized by functional category: computing capacity loss, data integrity consequences, physical infrastructure damage, and personnel safety consequences.

Computing capacity loss is characterized by percentage of facility IT load that becomes unavailable and the duration of unavailability. The severity spectrum runs from partial load reduction — less than twenty-five percent unavailable for less than one hour — through complete facility shutdown of extended duration exceeding seventy-two hours. The design basis acceptance criteria in Section 6 establish the recovery time objectives that bound each severity tier.

Data integrity consequences arise from non-graceful shutdown — abrupt loss of power or cooling without orderly shutdown sequencing — which can corrupt in-flight transactions, damage storage media, and cause loss of data that has not been replicated to a remote location. The design basis must establish the replication and journaling requirements that bound data integrity consequences under each postulated event category.

Physical infrastructure damage is characterized by the type and extent of equipment damage and the procurement and replacement timeline. High-voltage electrical equipment and custom cooling infrastructure have long procurement lead times that extend the consequence duration well beyond the initial disruption. Custom data center infrastructure components may have procurement timelines measured in months, a characteristic that becomes analytically significant under the PE-3 and PI-2 scenarios where physical restoration is the only recovery path.

Personnel safety consequences are addressed in the design basis primarily through the emergency response and site access provisions of the physical integrity event category. They are Tier 1 consequences but have Tier 2 and Tier 3 implications where emergency response is prevented or delayed by conditions outside the facility boundary.

5.2 Tier 2 — Grid-Level Consequences

Tier 2 consequences propagate from the facility to the bulk electric system. They are organized by the mechanism of propagation: load dropout effects, frequency perturbation, and voltage excursion.

Load dropout effects occur when a hyperscale facility's demand on the grid drops rapidly. Load dropout is the grid-side manifestation of both power supply events — where load drops because the facility has lost power and shifted to backup generation — and workload migration events — where load drops because the facility has actively migrated its computing workload to a remote location. The grid consequence of a large load dropout is an overfrequency condition in the affected area, as generation that was serving the departed load now has nowhere to deliver its output. Grid operators and their automatic response systems are calibrated primarily for underfrequency events. An overfrequency event of comparable magnitude receives less automatic compensating response and presents a different set of relay coordination challenges.

Frequency perturbation propagates from the point of the load dropout outward through the interconnection as a wave, at a speed governed by local inertia and network admittance. There is an interval between the initiating event and the arrival of the frequency perturbation at distant relay nodes — an interval during which human operators could theoretically intervene but during which situational awareness is limited because the originating event occurred inside a customer facility and is not directly observable in SCADA systems.

Voltage excursion consequences are secondary to the frequency effects for the workload migration scenario but are the primary consequence of power supply events that affect the transmission system near the facility. The April 7, 2015 Washington DC event documented a voltage excursion severe enough to cause Calvert Cliffs nuclear units 1 and 2 to initiate automatic shutdown — a Tier 3 consequence initiated by a Tier 2 voltage condition caused by a Tier 1 equipment failure at a substation serving the high-density load corridor.

5.3 Tier 3 — Cross-Sector Cascading Consequences

Tier 3 consequences are the formally defined innovation of this design basis. This tier accounts for consequences that propagate from the grid-level effects of a facility-level event into dependent critical infrastructure sectors and the services they support. Its inclusion reflects the Foundational Dependency characteristic of the hyperscale facility class established in Section 2.4: no other critical infrastructure facility has a dependency footprint that spans all other sectors simultaneously.

The cross-sector cascade mechanism operates through two pathways that are distinct in character and in their policy implications.

The digital dependency pathway operates through the services that the disrupted hyperscale facility was providing. When a hyperscale facility loses capacity, the workloads it was hosting — government operations, financial system transactions, military logistics, pipeline SCADA, grid control systems, communications infrastructure — lose their computing substrate. The consequences propagate through those dependent systems at rates determined by the dependent systems' own resilience characteristics: how long they can operate on cached data, how quickly they can fail over to alternative computing capacity, and whether alternative capacity is available and accessible.

The grid stress pathway operates through the bulk electric system effects documented in Tier 2. When Tier 2 consequences produce grid instability, the instability affects all loads connected to the affected portion of the grid. Nuclear plants in the affected area may initiate automatic shutdown on voltage protection. Industrial loads with undervoltage protection trip. Distribution system customers lose service.

The cross-interconnection consequence of the workload migration scenario belongs to this tier. When a forced compute migration drops load in the Eastern Interconnection and simultaneously surges load in ERCOT, both events are Tier 3 consequences of the original Tier 1 facility event. The causal chain crosses two regulatory boundaries — the facility meter and the interconnection seam — neither of which is currently monitored for this mechanism. NERC's jurisdiction stops at the customer meter. The IP orchestration layer that executes the compute migration crosses no boundary that any regulatory body currently monitors.

This regulatory observability gap is not merely a policy concern — it is an engineering design basis finding. A facility class whose failure modes can produce grid consequences in a separate interconnection through a mechanism that is invisible to all regulatory monitoring systems has a consequence taxonomy that no prior design basis has addressed. Formally including cross-sector cascading consequences in the design basis is the first step toward establishing the engineering requirements and acceptance criteria that would make those consequences plannable and regulatable.

Section 6 — Acceptance Criteria

6.1 Purpose of Acceptance Criteria

Acceptance criteria define what adequate means for each postulated event category. They establish the measurement standard against which actual facility capability is assessed in the gap analysis. Without acceptance criteria, a design basis produces a catalog of scenarios but no instrument for evaluating whether a facility meets or falls short of the design basis requirements.

The acceptance criteria in this section are expressed in three forms: operational state criteria — what condition must the facility be capable of reaching, and within what time — system performance criteria — what must specific facility systems be capable of demonstrating under postulated event conditions — and consequence bounding criteria — what consequence tier must be prevented by adequate facility response.

The acceptance criteria in this section are additive to, not replacements for, the Uptime Institute Tier Standard. The Uptime Institute's four-tier classification addresses topology and operational sustainability: power system redundancy, cooling infrastructure redundancy, concurrent maintainability, and operational continuity under normal and failure conditions. Tier IV — the Fault Tolerant classification — represents the highest level of the Tier Standard and is the minimum assumed baseline for hyperscale facilities evaluated under this design basis. A facility that does not meet Tier IV operational characteristics should address that shortfall before conducting the gap analysis in Section 7.

The Uptime Institute explicitly excludes security, regional weather, building codes, and property usage from its Tier Standards. That explicit exclusion defines the boundary between the Tier Standard's scope and this design basis. The acceptance criteria that follow address exactly the categories the Tier Standard excludes: extended and catastrophic-duration power supply events, workload migration events and their grid interaction consequences, physical security against deliberate attack, and cross-sector cascading consequences. A facility holding a current Tier IV certification satisfies the compliance floor for internal redundancy and availability. It does not satisfy the acceptance criteria of this design basis. The gap between Tier IV certification and this design basis is precisely the space where the capital investment identified by the gap analysis is concentrated.

6.2 Power Supply Acceptance Criteria

PE-1 (Short-duration utility power loss). Adequate facility response requires: no loss of computing capacity during the event; automatic transfer to backup power within the transfer switch design switching time, not to exceed two seconds for static transfer switches or ten seconds for mechanical transfer switches; and return to utility power supply without manual intervention when utility power is restored. Consequence bounding criterion: no Tier 2 or Tier 3 consequences from a short-duration power supply event.

PE-2 (Extended-duration utility power loss). Adequate facility response requires: sustained operation at not less than seventy-five percent of normal IT load for not less than seventy-two hours on backup generation; a documented and tested fuel resupply plan that specifically addresses area-wide disruption conditions — not only normal commercial fuel delivery logistics; and a load-shedding priority framework that identifies and preserves highest-consequence workloads — government, military, and financial sector operational technology — when fuel conservation is required. The fuel resupply plan must address denial-of-access conditions as a planning scenario, not as an excluded case. Consequence bounding criterion: no Tier 3 digital dependency pathway consequences arising from facility shutdown during an extended-duration power supply event.

PE-3 (Catastrophic-duration utility power loss). This event category does not carry an adequate facility response criterion in the conventional sense, because no current facility design can sustain full operation indefinitely without utility power. The acceptance criterion for PE-3 is a documentation and characterization requirement: the design basis requires that the facility's capability and limitation under this event be documented, that the consequence tier of facility shutdown under this condition be formally assessed and recorded, and that the gap between documented capability and the PE-3 condition be characterized as a finding that directly informs the military action extension analysis. Facilities are not expected to eliminate PE-3 consequences; they are required to know what those consequences are.

PE-4 (Voltage excursion without loss of power). Adequate facility response requires: a documented inventory of all behind-the-meter protective equipment with relay settings and trip thresholds specified; an assessment of the aggregate load dropout that would result if that equipment operated simultaneously in response to a voltage excursion; and confirmation that facility operations protocols do not set behind-the-meter protective equipment trip thresholds within the ANSI C84.1 Range A service voltage band without specific engineering justification. The load dropout magnitude assessment is required as direct input to the Tier 2 grid consequence analysis for PE-4.

6.3 Thermal Management Acceptance Criteria

TM-1 (Loss of mechanical cooling). Adequate facility response requires: documented thermal dwell time for the facility — the measured or analytically established interval from loss of primary mechanical cooling to the first protective temperature threshold activation for computing equipment, under worst-case ambient conditions and maximum IT load — of not less than fifteen minutes; a tested load-reduction sequence that extends thermal tolerance by at least fifty percent of the base thermal dwell time when activated; and a documented maximum ambient design temperature above which the cooling system cannot maintain equipment operating temperatures at full load, as a required input to environmental condition planning.

TM-2 (Loss of cooling water supply). Adequate facility response requires: not less than forty-eight hours of on-site cooling water storage capacity at planned full operating load under design ambient conditions; a documented and tested transition plan from water-cooled to air-cooled or reduced-capacity cooling operation, including the specific load reduction required to sustain operations under air-cooled conditions only; and an assessment of the load level at which air-cooled operation is thermally sustainable at the ninety-ninth percentile ambient temperature for the facility's climate zone.

TM-3 (Combined power and thermal stress). Adequate facility response requires: a documented coordinated load reduction protocol that specifies the sequence and rate of load reduction under simultaneous power supply degradation and cooling capacity reduction; demonstration that the protocol prevents thermal shutdown of any computing zone for at least thirty minutes while backup generation is operational; and confirmation that the protocol has been tested under simulated combined-stress conditions at a load level not less than seventy-five percent of normal IT load. The coordinated protocol must address the potential conflict between thermal management load reduction and backup power fuel conservation.

6.4 Workload Migration Acceptance Criteria

WM-1 and WM-2 (Workload migration events). The acceptance criteria for workload migration events depart from the conventional form because adequate facility response is not defined by whether the migration occurs — migration may be the correct operational decision — but by whether its grid consequences are anticipated, quantified, and communicated.

The design basis acceptance criterion for workload migration events is: the facility operator must have documented the maximum rate and magnitude of IT load dropout that can result from a workload migration event under both planned and emergency conditions; must have assessed the Tier 2 and Tier 3 grid consequences of that dropout for the relevant grid interconnection; and must have established a communication protocol with the applicable grid operator — PJM, MISO, ERCOT, or WECC — for advance notification of planned large-scale migrations. The notification protocol must specify a minimum notification lead time that allows the grid operator to prepare automatic generation control adjustments before the migration executes.

This criterion does not require the facility operator to obtain permission before migrating workloads, nor does it impose a constraint on migration decisions in response to an active threat. It requires that the grid consequence of migration be characterized and that the relevant grid operator have the information needed to prepare for the grid event the migration will produce.

WM-3 (Cross-interconnection load transient). The acceptance criterion for this scenario is at the cluster level, not the individual facility level. It requires that the applicable reliability coordinator — PJM for Northern Virginia — incorporate a coordinated workload migration event as a postulated stress scenario in its contingency analysis, and that this analysis be updated as cluster load grows. This criterion requires institutional engagement that is outside the scope of any individual facility operator. The engagement pathway is specified in Deferred Item DI-1 in Section 9.

6.5 Connectivity Acceptance Criteria

CE-1 (Single-provider connectivity loss). Adequate facility response requires: physical and logical diversity from not less than two independent internet service providers entering the facility campus via separate physical pathways and, where possible, separate geographic ingress points; automatic failover to the surviving provider path within sixty seconds of single-provider loss; and no customer-affecting connectivity outage resulting from CE-1.

CE-2 (Area-wide connectivity loss). The acceptance criterion for area-wide connectivity loss is a characterization and documentation requirement, parallel to the PE-3 acceptance criterion for power supply. The design basis requires that the facility document all physical fiber and cable ingress and egress routes, identify shared-corridor vulnerabilities where multiple provider paths share a common physical conduit or right-of-way, and assess the consequence tier of simultaneous loss of all connectivity paths. Facilities in the Northern Virginia cluster with fiber routes that share physical corridors must document that shared-corridor vulnerability as a specific finding. Area-wide connectivity loss is not a condition that commercial connectivity redundancy can prevent; it requires physical route diversity that not all facilities can achieve. The design basis requires that the limitation be known and documented.

6.6 Physical Integrity Acceptance Criteria

PI-1 (Fire). Adequate facility response requires: fire suppression systems compliant with NFPA 75 (Standard for the Fire Protection of Information Technology Equipment) and NFPA 2001 (Standard on Clean Agent Fire Extinguishing Systems) as applicable to the equipment types installed; a documented emergency response coordination arrangement with the local authority having jurisdiction that includes facility-specific hazard information — electrical equipment ratings, chemical agent inventories, battery system characteristics — required for safe fire response; and a documented restoration plan for fire-affected areas that includes equipment damage assessment, suppression agent cleanup, and reinstatement testing procedures.

PI-2 (Physical damage to critical infrastructure points). Adequate facility response requires documentation of all critical infrastructure points for the specific facility — at minimum: utility service entrance equipment, substation and primary distribution switchgear, backup generation plant, fuel storage system, and primary cooling plant — with the consequence tier of damage to each point assessed against the postulated event categories in Section 4. This critical infrastructure point characterization is the direct input to the military action extension analysis. Facilities cannot conduct a meaningful gap analysis under the military action extension without having first completed the PI-2 critical infrastructure point characterization at the baseline level.

Section 7 — Gap Analysis Framework

7.1 Purpose and Structure

The gap analysis is the instrument by which this design basis produces actionable outcomes. It translates the acceptance criteria of Section 6 into a structured assessment of a specific facility's capabilities, identifies the delta between demonstrated capability and required capability, and characterizes that delta in terms that support capital project identification, regulatory compliance planning, and insurance risk assessment.

The gap analysis is conducted for each postulated event category in turn. For each category, the assessor documents the facility's demonstrated capability against the relevant acceptance criterion, characterizes any shortfall, assigns a severity rating, and identifies the engineering measures available to close the gap.

7.2 Gap Severity Classification

Gaps are classified in three severity tiers that reflect the consequence implications of the shortfall.

Class	Consequence Tier	Definition
Class A	Tier 3 Exposure	Gap creates conditions under which a postulated event produces cross-sector cascading consequences. Highest priority for remediation. WM-1 and WM-2 documentation and notification gaps are Class A by definition.
Class B	Tier 2 Exposure	Gap creates conditions under which a postulated event produces grid-level consequences within a manageable envelope.

Class	Consequence Tier	Definition
Class C	Tier 1 Exposure	Gap creates conditions under which a postulated event produces facility-level consequences contained within the facility boundary.

7.3 The Gap as Capital Investment Instrument

The gap analysis is the data center sector's equivalent of the finding-to-license-condition pathway that drives capital investment in nuclear facilities. When a nuclear plant's design basis analysis identifies a gap, that finding drives a license condition and a capital project. The pathway from analytical finding to funded engineering work is institutionalized and mandatory.

For the data center sector, the mandatory pathway does not yet exist. The regulatory gap documented in the DBA-DC Policy Companion means that the gap analysis findings from this design basis are not currently enforceable. They are, however, immediately actionable by three categories of actor who do not require regulatory mandate.

Asset owners with government and military cloud service customers face contractual and reputational consequences if their facilities cannot demonstrate design basis adequacy. The gap analysis provides the documentation needed to support both the facility assessment and the remediation planning.

Insurers underwriting data center physical risk face uncharacterized exposure to Tier 2 and Tier 3 consequences that current actuarial models do not account for. The gap analysis provides the structured risk characterization that actuarial assessment requires.

Regulators — FERC, NERC, and the emerging data center regulatory framework — will need a standard against which to measure compliance when mandatory standards are established. The gap analysis framework in this document is designed to be that standard. Its adoption as a mandatory requirement is a policy decision; its analytical content is complete and ready for that purpose.

7.4 Tier Certification as the Gap Analysis Starting Point

The Uptime Institute's Tier Certification program verifies a facility's design and operational compliance against the Tier Standard specifications through independent assessment. Many hyperscale facilities hold current Tier III or Tier IV certifications. That certification record is the appropriate starting point for a gap analysis against this design basis, not a redundant one.

A gap analysis that begins from a verified Tier IV baseline does not need to reconstruct the power system redundancy, cooling infrastructure redundancy, and concurrent maintainability assessments that Tier certification already documents. Those findings are accepted as given. The gap analysis then works the delta: the postulated event categories and acceptance criteria in this design basis that the Tier Standard explicitly does not address.

For a Tier IV certified facility, that delta is well-defined. The extended and catastrophic-duration power supply events test capability beyond the availability focus of the Tier Standard — particularly the fuel resupply and denial-of-access conditions the Tier Standard assumes away. The workload migration event category has no analog in the Tier Standard, which is scoped to the facility boundary. The PI-2 critical infrastructure point characterization is outside the Tier Standard's scope. All Tier 2 and Tier 3 acceptance criteria address consequences that propagate beyond the facility boundary and are therefore entirely outside the Tier Standard's analytical frame. These are the categories where gap findings will concentrate and where capital investment will be required.

For a facility that has not achieved Tier IV certification, the gap analysis should note the Tier shortfall as a prerequisite finding and address it separately before proceeding to the design basis gap analysis. The Uptime Institute Tier Certification and this design basis gap analysis are complementary instruments covering adjacent but non-overlapping scopes. Together they constitute the complete structured assessment of a hyperscale facility's capability against both its internal availability requirements and its external consequence obligations.

Section 8 — Relationship to the Military Action Extension

8.1 The Bridge Between Design Basis and Military Action Analysis

This design basis establishes the envelope of conditions the hyperscale data center facility class must be demonstrated to withstand. The postulated events in Section 4 are defined without reference to cause: PE-3 addresses catastrophic-duration utility power loss regardless of whether that loss results from equipment failure, severe weather, or deliberate physical attack. The design basis is cause-agnostic by design.

The military action extension is cause-specific. It asks: when military action is the initiating condition for the events defined in this design basis, what additional analytical requirements does the cause introduce? The answer is structured around the four DBA-MA differentiators established in the parent DBA-MA framework.

8.2 The Four DBA-MA Differentiators Applied to Data Center Facilities

Intentionality. A military actor targeting a hyperscale data center is not subject to the probabilistic independence assumptions that underlie normal reliability analysis. The actor selects which facility to target, which infrastructure points within the facility to attack, and what timing and sequencing to apply. The design basis gap analysis findings — particularly the critical infrastructure point characterization required by PI-2 — directly inform the military actor's targeting problem. The military action extension must analyze the design basis gap findings from the perspective of an adversary who has read this document and who will attack the identified gaps in preference to hardened systems.

Persistence. A military actor can maintain pressure on a facility and its supporting infrastructure over an extended period, preventing the resupply and repair activities that the PE-2 and PE-3 acceptance criteria assume. The seventy-two-hour fuel resupply plan required by the PE-2 acceptance criterion may be rendered ineffective if the road network, fuel supply chain, or repair crews are denied access by ongoing military activity. The military action extension must assess each PE-2 and PE-3 mitigation measure against a persistent denial-of-access condition — not merely against the normal commercial disruption conditions that informed the baseline planning.

Precision. A military actor using precision-guided munitions can target the critical infrastructure points identified in the PI-2 characterization with a reliability that makes the independent failure assumptions of normal reliability analysis inapplicable. Redundant backup generation plants, separated by the distance required to prevent common-cause failure from equipment malfunction, may not provide equivalent protection against a precision strike that can engage both plants in the same engagement window. The March and April 2026 Iranian Shahed drone strikes on AWS and Oracle data center facilities in the UAE and Bahrain demonstrated precisely this dynamic: the adversary directed autonomous platforms to specific facility locations, not to a geographic area.

Denial. Military action can prevent access to the facility for inspection, maintenance, repair, and emergency response. Every acceptance criterion in Section 6 that involves a restoration timeline assumes that personnel can reach the facility and begin work. Denial of site access — through physical control of access routes, area denial weapons, or occupation — invalidates those timeline assumptions and extends every Tier 1 consequence into the Tier 2 and Tier 3 ranges. The four DC series distinguishing characteristics compound this: foreign ownership creates a pre-positioning vector through which denial may be achieved from within the facility before external denial is established.

8.3 The Evidentiary Foundation for the Military Action Extension

The military action extension for the data center sector has its evidentiary foundation in the March and April 2026 Iranian Shahed drone strikes on Amazon Web Services facilities in the UAE and Bahrain, and on the Oracle data center in Dubai. These were the first confirmed instances of a state adversary deliberately targeting commercial data center infrastructure as an act of war. The strikes demonstrated all four DBA-MA differentiators in operation: the adversary selected specific facilities from among many available targets; the campaign extended over five weeks with multiple strikes; the Shahed drone platform was directed to specific facility locations; and the strikes were intended to deny the facilities' customers the computing services they depended on.

This evidentiary foundation is to the DBA-DC military action extension what the Russian military seizure and sustained occupation of the Zaporizhzhia nuclear power plant is to the DBA-MA nuclear series: the real-world demonstration that the gap the extension addresses is not theoretical. The design basis in this document is the analytical instrument needed before the extension can be rigorously applied. The extension follows from this foundation.

8.4 Sequence of the Military Action Extension Analysis

The military action extension analysis proceeds from this design basis in a defined sequence.

First, the critical infrastructure point characterization required by PI-2 must be completed. The extension cannot be conducted without knowing which facility systems are the priority targets under an intentional attack.

Second, each PE and TM acceptance criterion is re-evaluated under persistence and denial conditions. The question is not whether the facility meets the baseline criterion — that is the design basis finding — but whether the mitigation measures that allow it to meet the baseline criterion remain effective when access is denied and resupply is interdicted.

Third, the Concentration characteristic amplifies every finding at the cluster level. A military actor who has read this design basis and the PI-2 characterization for multiple Northern Virginia campuses can construct a coordinated attack scenario that exploits the shared-infrastructure vulnerabilities identified in Section 2.4. The military action extension must be conducted at the cluster level, not only at the individual facility level, to capture this amplification.

Fourth, the Foreign Ownership characteristic requires that the military action extension explicitly consider the pre-positioning scenario: the possibility that foreign-affiliated ownership or operation of facility infrastructure has created adversary access that does not require external kinetic attack to activate. DBA-DC-L3-003 develops this scenario in full. The design basis establishes the analytical category; the extension develops the consequence chain.

Section 9 — Deferred Items and Future Development

9.1 Deferred Items

The following items are formally deferred from this version of the design basis. Deferred items are distinguished from open items in that they represent requirements whose fulfillment depends on external engagement or institutional processes that cannot be completed within the analytical scope of this document. Each deferred item specifies the scope of what is deferred, the resolution pathway, and the resolution criterion that will allow the item to be closed at a subsequent revision pass.

Deferred Item	Scope	Resolution Pathway
DI-1: WM-3 NERC Contingency Engagement	The WM-3 cross-interconnection load transient acceptance criterion requires that the applicable reliability coordinator	Engagement to be initiated through the NERC disturbance analysis process, using the evidentiary record established

Deferred Item	Scope	Resolution Pathway
	incorporate a coordinated workload migration event as a postulated stress scenario in contingency analysis. This requirement cannot be fulfilled by this document alone; it requires engagement with NERC's reliability planning process and, specifically, with PJM Interconnection as the reliability coordinator for the Northern Virginia cluster.	by the September 2007 MRO disturbance and the April 2015 Washington DC area event as the technical foundation. Bob Cummings (former NERC, IEEE) is the identified bridge contact. Resolution criterion: PJM confirms incorporation of a coordinated workload migration scenario in its Regional Transmission Expansion Planning (RTEP) contingency analysis, or formally documents its assessment of why the scenario falls outside its planning scope.
DI-2: Facility-Level Gap Analysis Validation	The acceptance criteria in Section 6 require validation against one or more actual hyperscale campuses to confirm that the postulated event categories and criteria reflect real facility architectures and operational conditions. No facility-level validation has been conducted at this version.	Engagement to be pursued with a hyperscale operator willing to conduct a confidential gap analysis against this design basis. The Northern Virginia cluster is the priority validation environment. Resolution criterion: gap analysis completed for at least one campus exceeding 200 MW IT load, with findings incorporated into a Section 6 errata or Section 7.5 validation annex at the next revision pass.
DI-3: Quantitative Calibration of Bounded Thresholds	Several acceptance criteria in Section 6 establish bounded thresholds (voltage tolerance ranges, thermal dwell times, water storage durations) that reflect engineering judgment at this version but require calibration against actual facility instrumentation data and grid operator planning parameters.	Calibration to be conducted in conjunction with DI-2 facility-level validation. Grid operator calibration (PE-4 voltage thresholds in operational context, WM-3 magnitude thresholds for PJM planning purposes) to be pursued in conjunction with DI-1. Resolution criterion: updated quantitative thresholds documented and accepted by technical reviewers at next revision pass.

9.2 Future Development

The following directions are anticipated for subsequent revisions of this document and for related series documents.

Downstream facility-class design bases. The DBA-ES parent document will emerge inductively as facility-class design bases are developed for large hydroelectric facilities, HVDC interconnects, and major natural gas compressor stations. The methodology layer in Section 3 of this document is the seed of that parent. Each subsequent facility-class design base should be compared against Section 3 to refine the transferable methodology and identify where facility-specific adaptation is required.

DBA-DC military action extension. The military action extension for the data center facility class is the next document in this series following technical review and initial revision of this document. It inherits from this design basis and applies the four DBA-MA differentiators to each postulated event category. The extension must be conducted at both the individual facility and cluster levels. Section 8.4 establishes the sequence of that analysis.

DBA-DC-UxS sector implementation annex. DBA-UxS (v1.0), the cross-sector foundational document for uncrewed systems threats to critical infrastructure, establishes the requirement for a DBA-DC sector implementation annex. That annex — to address UAS, USV, and UUV threat vectors as they apply to hyperscale data center facilities — is in development.

Lower-consequence facility tier extension. The design basis for colocation, enterprise, and edge data center facility tiers is anticipated as a subsequent extension of this framework. The threshold definition in Section 2.1 and the consequence taxonomy in Section 5 establish the boundary conditions for that extension.

International cluster annexes. The methodology of this design basis is transferable to allied-nation hyperscale clusters without modification. Jurisdictional annexes addressing the UAE and Bahrain cluster — the location of the March 2026 Shahed strikes — and the United Kingdom and Western European clusters are anticipated.

Section 10 — Facility Class Instruments

10.1 The Inheritance Model

Facility-class documents in the DBA-DC series inherit the design basis methodology, postulated event categories, consequence taxonomy, acceptance criteria framework, and gap analysis structure established in this document. They do not repeat this material. Each facility-class document adds what is specific to its tier profile: facility-tier-specific postulated event parameters, consequence thresholds, acceptance criteria calibrations, and gap analysis findings that this document establishes in general terms but does not develop for each tier. The four distinguishing characteristics established in Section 2.4 govern every document in the DBA-DC series.

10.2 Facility Class Register

The following table registers the facility classes addressed by the DBA-DC series. It identifies the current development status of each facility class and the document or documents issued or anticipated for it. The register is the instrument by which the completeness of the series' coverage can be assessed at a given point in time.

Facility Class	**FC Designation**	**Current Document**	**Status**
Hyperscale Data Center (>100 MW IT Load)	FC1	DBA-DC-FC1 v1.0 (this document)	Active
Colocation Data Center	FC2	None	Not Yet Addressed
Enterprise Data Center	FC3	None	Not Yet Addressed
Edge Data Center	FC4	None	Not Yet Addressed